Program 7: Basics of Ansible: Inventory, Playbooks, and Modules, Automating Server Configurations with Playbooks, Hands-On: Writing and Running a Basic Playbook

Inventory:

In Ansible, the inventory is a file that contains a list of target machines (hosts) where automation tasks will be executed. It defines groups or individual hosts with connection details. For example, using localhost **ansible_connection=local** allows you to test and run playbooks on your own system without needing remote access. The inventory can be static **(INI/YAML format)** or dynamic, depending on your infrastructure.

Playbook:

A playbook in Ansible is a YAML-formatted file that outlines the desired state of a system through a series of tasks. Each play defines which hosts to target and what actions to perform using Ansible modules. Playbooks are human-readable, version-controllable, and reusable, making them ideal for automating server setup, software installation, configuration changes, and more.

Module:

Modules are the core building blocks in Ansible used to carry out specific operations within tasks. They are like small programs that perform actions such as installing packages **(apt, yum)**, managing files **(copy, file)**, controlling services **(service)**, or providing debug output **(debug)**. Modules make Ansible highly extensible and easy to use for a wide range of automation tasks.

Step 1: Install Ansible on Ubuntu

```
sudo apt update
sudo apt install ansible -y
```

Step 2: Check Ansible is Installed

```
ansible –version
```

Step 3: Test Connection with localhost - Create or edit the Ansible inventory file

```
sudo nano /etc/ansible/hosts
```

Add:

```
[local]
localhost ansible_connection=local
```

Then test:

```
ansible localhost -m ping
```

(You should see a "**pong**" response indicating success)

```
[root@centos01 ~]# ansible localhost -m ping
localhost | SUCCESS => {
    "changed": false,
    "ping": "pong"
```

Step 4: Create a Directory for Playbooks

```
mkdir ~/ansible-playbooks

cd ~/ansible-playbooks
```

Step 5: Create First Playbook File - Create the file firstpb.yml

```
nano firstpb.yml
```

Code:

```
---
- name: First Basic PB
  hosts: localhost

  tasks:
  - name: Test Connectivity
    ping:
  - name: "Print Output"
    debug: msg="Alright!!"
```

After writing the code, save the file and exit the editor.

Step 6: Run the Playbook

```
ansible-playbook firstpb.yml
```

You should see both:

✅ **ping success message**

✅ **debug output: "Alright!!"**

```
ok: [localhost]

TASK [Test Connectivity] ***
**********
ok: [localhost]

TASK [Print Output] ********
**********
ok: [localhost] => {
    "msg": "Alright!!"
}
```